Exercise Set 3:

Ex 1:

$$\vec{F_T} = \vec{F_1} + \vec{F_2} + \vec{F_3}$$

$$F_{Tx} = F_{1x} + F_{2x} + F_{3x}$$

$$F_{Ty} = F_{1y} + F_{2y} + F_{3y}.$$

$$\vec{F_T} = (-38)\vec{i} + (-1)\vec{j}.(N)$$

1/ Direction of a vector:

معرفة الزاوية θ التي يميل بها عن الأفق (Ox).

$$a_x = a\cos\theta$$
$$a_y = a\sin\theta.$$

$$\frac{a_y}{a_x} = \tan(\theta)$$

$$\theta = \tan^{-1}\left(\frac{a_y}{a_x}\right).$$

$$= \tan^{-1}\left(\frac{\frac{F_y}{\cancel{m}}}{\frac{F_x}{\cancel{m}}}\right) = \tan^{-1}\left(\frac{F_y}{F_x}\right)$$

$$\theta = \tan^{-1}\left(\frac{1}{38}\right) = 1{,}5$$

2/

$$m = \frac{F_T}{a} = \frac{\sqrt{1^2 + (-38)^2}}{3{,}75} = \text{---}$$

3/

$$\vec{a} = \frac{d\vec{V}(t)}{dt}.$$

$$\vec{a}\,.dt = d\vec{V}(t)$$

$$\begin{cases} V_x = a_x t + V_{0x} \\ V_y = a_y t + V_{0y} \end{cases}$$

$$\begin{cases} V_x = a\cos\theta\, t \\ V_y = a\sin\theta\, t. \end{cases}$$

Ex02:

1/

$$F = m_1 a_1 = m_2 . a_2$$

$$\frac{m_1}{m_2} = \frac{a_1}{a_2} = \frac{1}{3}$$

2/

$$F = m_1 a_1 = m_2 a_2$$

$$= (m_1 + m_2) a_3.$$

$m_2 = 3 m_1$ so:

$$F = 4 m_1 a_3.$$

$$F = m_1 a_1$$

$$4 \cancel{m_1} a_3 = \cancel{m_1} a_1$$

$$a_3 = \frac{a_1}{4} = \frac{3}{4} \, m/s^2.$$

Ex03:

1

$$v(t) = a(t - t_0) + v_0(t_0)$$

$$v = at$$

$$a = \frac{v}{t} = \frac{v}{4t}$$

CamScanner

frictionless = No احتكاك

$$x = \frac{1}{2}a(t-t_0) + v_0.(t-t_0) + x_0(t_0)$$

$$d = \frac{1}{2}at^2$$

$$d = \frac{1}{2}\left(\frac{v}{\Delta t}\right)(\Delta t)^2 = \boxed{\frac{1}{2}v.\Delta t}$$

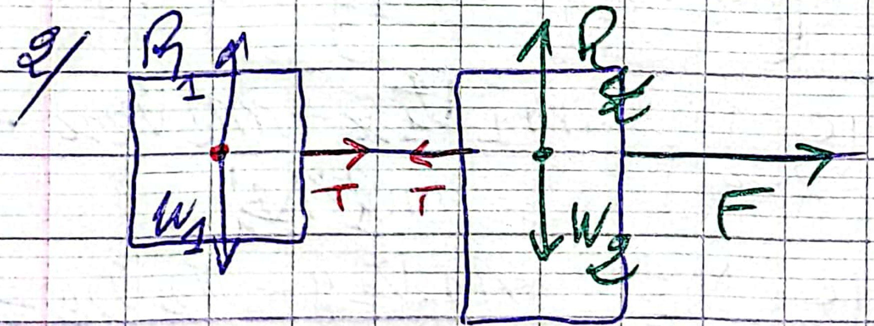
2) $$\boxed{F = ma = m\frac{v}{\Delta t}}$$

Ex04:

1/

(لا يتمدد) The rope is inextensible implies that: $a_1 = a_2 = 2,9\,m/s^2$

2/



$$T = m a_1 = 4 \times 2,9\,N$$

3/

$$\vec{F} + \vec{T} = m\vec{a_2} \qquad |\vec{F}_{net} = m.\vec{a_2}$$

$$F - T = m\,a_2$$

$$F = T + ma_2 \Rightarrow F > T$$

CamScanner

Exo5 :

1.

$$\vec{P_b} = \vec{P_a}$$

$$\vec{P_1} + \vec{P_2} = \vec{P_1'} + \vec{P_2'}.$$

$$\begin{cases} P_{1x} + P_{2x} = P'_{1x} + P'_{2x} \\ P_{1y} + P_{2y} = P'_{1y} + P'_{2y}. \end{cases}$$

$$\begin{cases} m_1 \cdot v_{1x} + m_2 v_{2x} = m_1 v'_{1x} + m_2 v'_{2x}. \\ m_1 \cdot v_{1y} + m_2 v_{2y} = m_1 v'_{1y} + m_2 v'_{2y} \end{cases}$$

$$v'_{2x} = \frac{m_1 v_{1x} + m_2 v_{2x} - m_1 v'_{1x}}{m_2}$$

$$v'_{2y} = \frac{m_1 v_{1y} + m_2 v_{2y} - m_1 v'_{1y}}{m_2}.$$

$$\begin{cases} v_{1x} = v_1 \cos(\theta_1) \\ v_{1y} = -v_1 \sin(\theta_1) \end{cases}$$

$$\begin{cases} v_{2x} = -v_2 \sin(\theta_2). \\ v_{2y} = v_2 \cos(\theta_1). \end{cases}$$

الحالتين

مفتوح 1

2– $\vec{P}_b = \vec{P}_a$

$$(m_1+m_2)\vec{v} = m_1\vec{v}_1 + m_2\vec{v}_2$$

$$v_x = \frac{m_1 \cdot v_{1x} + m_2 v_{2x}}{m_1+m_2}$$

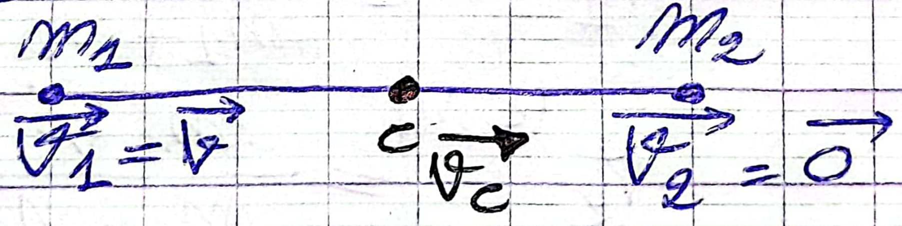
$$v_y = \frac{m_1 \cdot v_{2x} + m_2 v_{2x}}{m_1+m_2}$$

$$\vec{v} = v_x\vec{i} + v_y\vec{j}$$

$$v = \sqrt{v_x^2 + v_y^2}$$

Ex 06 Stationary = ثابت

1–



$$\vec{r}_C = \frac{m_1 \vec{r}_1 + m_2 \vec{r}_2}{m_1 + m_2}$$

$$\vec{v}_C = \frac{m_1 \vec{v}}{m_1 + m_2}.$$

2 –

In the C.M frame:

$\vec{P}_T = \vec{0}$. في الـ (CM).

دساتير تحويل المعلم

$$\begin{cases} x = X + x_C \\ y = Y + y_C \end{cases}$$

البرهان:

$$\vec{OM} = \vec{OC} + \vec{CM}$$

$$\Downarrow \quad \Downarrow \quad \Downarrow$$

$$\begin{pmatrix} x \\ y \end{pmatrix} \begin{pmatrix} x_C \\ y_C \end{pmatrix} \begin{pmatrix} X - 0 \\ Y - 0 \end{pmatrix}$$

$$\vec{v}_1 = \vec{v} = \vec{v}_1^{(CM)} + \vec{v}_C$$

$$\vec{v}_1^{(CM)} = \vec{v} - \vec{v}_C = -\frac{m_1 \vec{v}}{m_1 + m_2} + \vec{v}$$

$$= \frac{m_2 \vec{v}}{m_1 + m_2}.$$

$$\vec{v}_2 = \vec{0} = \vec{v}_2^{(CM)} + \vec{v}_C$$

$$\vec{v}_2^{(CM)} = -\vec{v}_C$$

$$= -m_1 \vec{v} / (m_1 + m_2)$$

So,

$$v_1^{(CM)} = \frac{m_2 v}{m_1 + m_2}$$

$$v_2^{(CM)} = \frac{m_1 v}{m_1 + m_2}$$

3 -

$$\vec{P}_{CM}(\text{before}) = \vec{0} = \vec{P}_1^{(CM)} + \vec{P}_2^{(CM)}$$

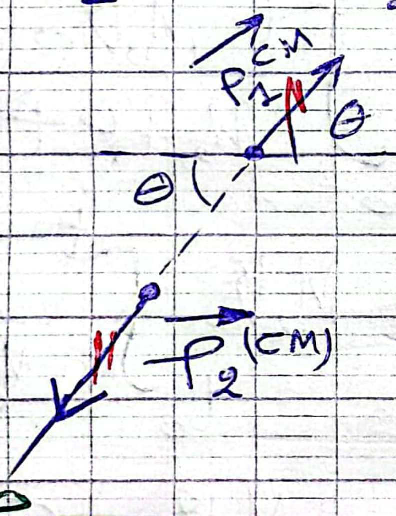
and

$$\vec{P}'_{CM}(\text{after}) = \vec{0} = \vec{P'}_1^{(CM)} + \vec{P'}_2^{(CM)}$$

$$\vec{P}_1^{(CM)} = m_1 \vec{V}_1^{(CM)} = \frac{m_1 \cdot m_2 \cdot \vec{V}}{m_1 + m_2}$$

$$\vec{P}_2^{(CM)} = \frac{-m_1 m_2 \vec{V}}{m_1 + m_2}$$

$$\vec{P}_1^{(CM)} = -\vec{P}_2^{CM}$$



Exo 07:

$$\sum \vec{F}_{ext} = \vec{W} + \vec{R} = \vec{0}$$

$$\frac{d\vec{P}}{dt} = \vec{0}$$

$\vec{P}$ is constant

$$P(0) = P(t)$$

$$M_0 V_0 = M V(t)$$
$$= \underbrace{M(t)}_{M_0 + K \times t} V(t)$$

$$M_0 V_0 = (M_0 + K \cdot t) V(t)$$

$$V = \frac{M_0 V_0}{M_0 + K t}$$

$$\frac{dx}{dt} = \frac{M_0 V_0}{M_0 + K t}$$

$$(x_0 = 0) \quad x = \int_{t_0}^{t} \frac{M_0 V_0}{M_0 + K t} \, dt$$

$$= M_0 V_0 \int_{t_0}^{t} \frac{1}{M_0 + K t}$$

$$= \frac{M_0 V_0}{K} \int_{t_0}^{t} \frac{1}{\frac{M_0}{K} + t}$$

$$= \frac{M_0 \cdot V_0}{K} \left[\ln \left(\frac{M_0}{K} + t \right) \right]_{t_0}^{t}$$

$$= \frac{M_0 \cdot V_0}{K} \left(\ln \left(\frac{M_0}{K} + t \right) - \ln \left(\frac{M}{K} \right. \right.$$

$$= \frac{M_0 \cdot V_0}{k} \ln\left(1+\frac{Mt}{k}\right)$$

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Exc 6. (ت ع)

4-

$$\vec{V_1'} = \vec{V_1'^{(CM)}} + \vec{V_c}$$

$$V_{1x}' = \|\vec{V_1'^{(CM)}}\| \cos\theta + \frac{m_1 V}{m_1+m_2}$$

$$V_{1y}' = \|\vec{V_1'^{(CM)}}\| \sin\theta + 0$$

$$\|\vec{V_1'}\| = \sqrt{(V_{1x}')^2 + (V_{1y}')^2}$$

$$\|\vec{V_1'^{(CM)}}\| = \frac{m_2 V}{m_1+m_2}$$

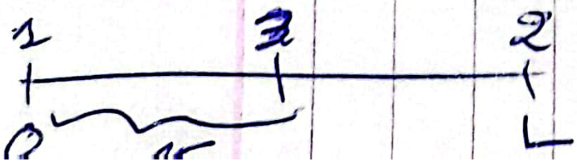
$$\|\vec{V_1'}\| = \frac{V}{m_1+m_2}\sqrt{m_1^2+m_2^2+2m_1m_2\cos\theta}$$

Exc 8.

1/ $\vec{F}_{1/3} + \vec{F}_{2/3} = \vec{0}$

$$-\vec{F}_{2/3} = \vec{F}_{1/3}$$

$$F_{2/3} = F_{1/3} \Rightarrow \cancel{G}\frac{2\cancel{m}\cdot M}{(L-x)^2} = \cancel{G}\frac{\cancel{m}\cdot \cancel{M}}{x^2}$$



حيث: $x \leq L$

$$\frac{2}{(L-x)^2} = \frac{1}{x^2}.$$

$$x^2 = \frac{(L-x)^2}{2}.$$

$$x = \frac{|L-x|}{\sqrt{2}}$$

$$\sqrt{2}\,x = L-x$$

$$x(\sqrt{2}+1) = L$$

$$x = \frac{L}{1+\sqrt{2}}.$$

2/
$$\vec{F}_{net} = \vec{F}_{1/3} + \vec{F}_{2/3}$$

($\vec{F}_{net} = \vec{0}$: حالة خاصة)

(إذا كان ③ بين ② و ①)

Exc 9 :

1/ $$a_N = \frac{v^2}{r_1} = \frac{(2\pi r_1)^2}{(T_1)^2} \times \frac{1}{r_1}$$

$$= \frac{4\pi^2 r_1}{T_1^2}$$

2/

$$a_{N_2} = \frac{4\pi^2 \cdot r_2}{T_2^2}$$

3/

$$F_1 = \frac{G \cdot M_T \cdot \cancel{m_m}}{r_1^2} = \cancel{m_m} \cdot a_1$$

$$F_2 = \frac{G \cdot M_J \cdot \cancel{m_S}}{r_2^2} = \cancel{m_S} \cdot a_2$$

$$\frac{F_2}{F_1} = \frac{a_2}{a_1} = \frac{\frac{M_J}{r_2^2}}{\frac{M_T}{r_1^2}}$$

$$\Rightarrow \frac{M_J}{M_T} = \frac{r_2^2 \cdot a_2}{r_1^2 \cdot a_1}$$